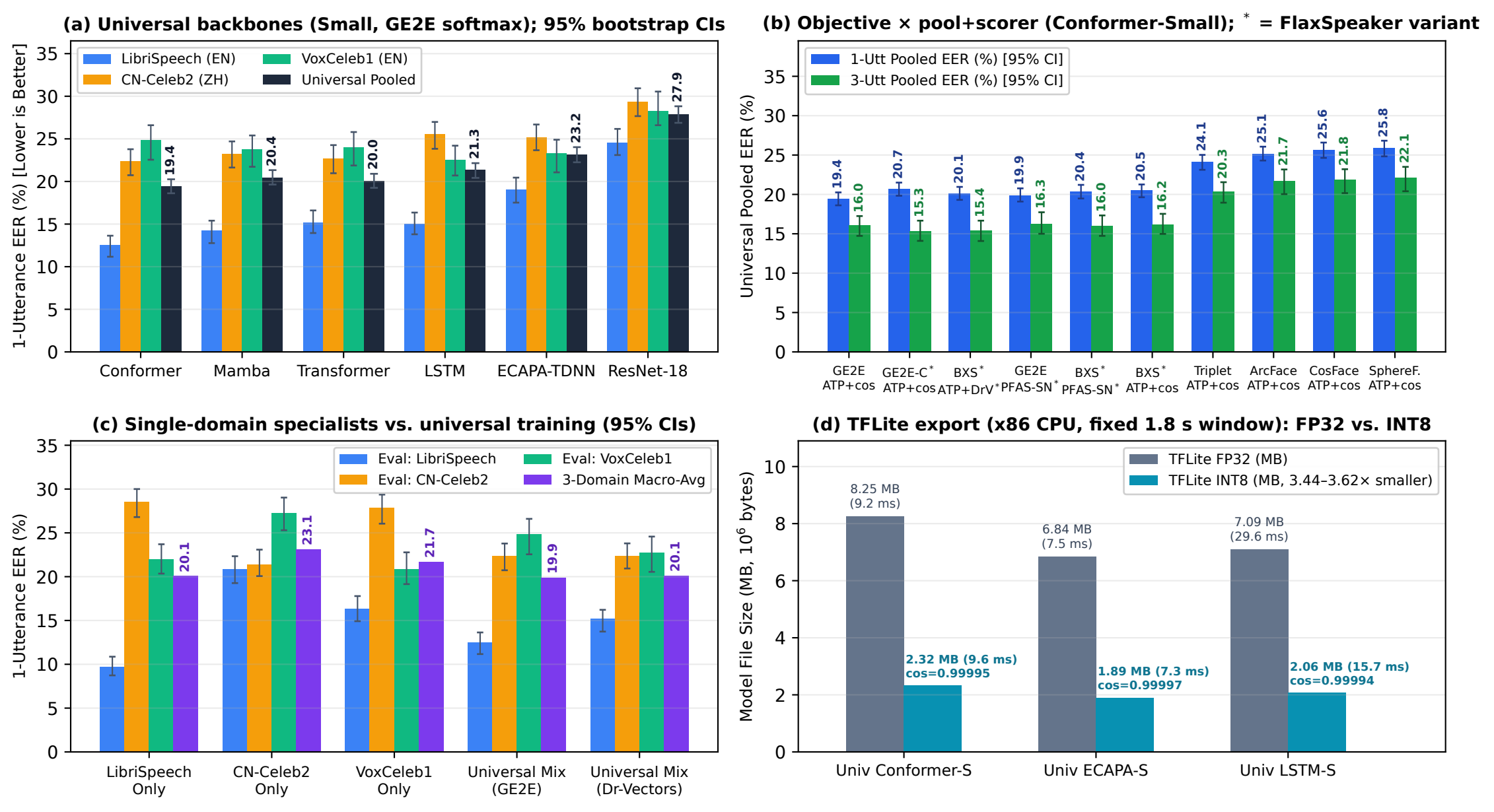
(a) Universal backbones (Small, GE2E softmax); 95% bootstrap CIs
1-Utterance EER (%) [Lower is Better]
LibriSpeech (EN)
CN-Celeb2 (ZH)
VoxCeleb1 (EN)
Universal Pooled
19.4
20.4
20.0
21.3
23.2
27.9
Conformer
Mamba
Transformer
LSTM
ECAPA-TDNN
ResNet-18
(b) Objective × pool+scorer (Conformer-Small); * = FlaxSpeaker variant
Universal Pooled EER (%)
1-Utt Pooled EER (%) [95% CI]
3-Utt Pooled EER (%) [95% CI]
19.4
16.0
20.7
15.3
20.1
15.4
19.9
16.3
20.4
16.0
20.5
16.2
24.1
20.3
25.1
21.7
25.6
21.8
25.8
22.1
GE2E ATP+cos
GE2E-C* ATP+cos
BXS* ATP+DrV*
GE2E PFAS-SN*
BXS* PFAS-SN*
BXS* ATP+cos
Triplet ATP+cos
ArcFace ATP+cos
CosFace ATP+cos
SphereF. ATP+cos
(c) Single-domain specialists vs. universal training (95% CIs)
1-Utterance EER (%)
Eval: LibriSpeech
Eval: CN-Celeb2
Eval: VoxCeleb1
3-Domain Macro-Avg
20.1
23.1
21.7
19.9
20.1
LibriSpeech Only
CN-Celeb2 Only
VoxCeleb1 Only
Universal Mix (GE2E)
Universal Mix (Dr-Vectors)
(d) TFLite export (x86 CPU, fixed 1.8 s window): FP32 vs. INT8
Model File Size (MB, 10^6 bytes)
TFLite FP32 (MB)
TFLite INT8 (MB, 3.44–3.62× smaller)
8.25 MB (9.2 ms)
2.32 MB (9.6 ms) cos=0.99995
6.84 MB (7.5 ms)
1.89 MB (7.3 ms) cos=0.99997
7.09 MB (29.6 ms)
2.06 MB (15.7 ms) cos=0.99994
Univ Conformer-S
Univ ECAPA-S
Univ LSTM-S